

Yuriy Tsimashenka

C++ Software Engineer — High-Performance Backend Systems

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SUMMARY

C++ engineer focused on high-performance, multithreaded network services and performance-critical systems. Hands-on experience with lock-stripped concurrent data structures, epoll-based event-driven servers, write-ahead logging, and SQL-backed persistence (PostgreSQL, SQLite), alongside REST API development in Python and JVM frameworks. Works independently in a fully remote setting, taking projects from design and implementation through CI/CD and deployment, and integrating with external services and partner systems.

CORE SKILLS

C++ & Concurrency: C++, Multithreading, Lock-Striped Data Structures, Atomic-Based Synchronization, Event-Driven Architecture

Systems & Networking: Linux epoll, TCP/IP, IPv4, UDP, Custom Network Protocols, Write-Ahead Logging, Fault-Tolerant Design

Databases & APIs: SQL, PostgreSQL, SQLite, REST APIs (Python and JVM frameworks), Backend Development

Other Languages: Python; English

Tools: Docker, Git, CMake, GitHub Actions, GitLab CI/CD, GCC, GDB

EXPERIENCE

Software Engineer — Contract / Freelance

Mar 2024 – Present

Remote

- Design and implement high-performance backend services and networked components in **C++** (and Python where suited), delivering reliable data-exchange workflows and interoperability across connected systems.
- Build **multithreaded, event-driven** systems for high-concurrency workloads, using asynchronous communication patterns to handle large volumes of simultaneous connections.
- Engineer fault-tolerant components with resilient error handling, recovery strategies, and restart-safe state management for business-critical services.
- Develop **RESTful interfaces** in Python and JVM frameworks and work with **PostgreSQL** to integrate with external services and partner systems.
- Own development and deployment workflows using Docker, Git, CMake, and CI/CD pipelines for consistent, repeatable releases.

SELECTED PROJECTS

CoreKV — Multithreaded In-Memory Key-Value Database

- Architected a multithreaded key-value server in **C++** with a custom TCP text protocol supporting concurrent CRUD operations.
- Built an event-driven network layer on **Linux epoll** to manage large numbers of concurrent connections.
- Implemented a **lock-stripped concurrent hash map** to reduce contention between readers and writers, and **Write-Ahead Logging** for persistence and crash recovery across restarts.

C++, Linux epoll, Multithreading, TCP, Write-Ahead Logging, Event-Driven Architecture

Network Packet Sniffer & Traffic Analyzer

- Built a multithreaded packet capture and analysis application in **C++**, with thread-safe communication between capture and analysis components using **atomic-based synchronization**.
- Parsed IPv4, TCP, and UDP traffic and integrated **SQLite** for storing analyzed traffic data.

C++, CMake, TCP/IP, IPv4, UDP, Multithreading, SQLite

XENOCIDE — Low-Level 2D Engine

- Developed a low-level 2D rendering engine in **C++** with custom abstractions for shaders, textures, and OpenGL resources.
- Implemented **batch rendering and optimization** for scenes with thousands of active entities; integrated Dear ImGui for debugging, level editing, and runtime tooling.

C++, OpenGL, GLFW, GLAD, ImGui

EDUCATION

Bachelor's Degree in Computer Software Engineering

Aug 2024 – Aug 2028 (expected)

Belarusian State University of Informatics and Radioelectronics